Zilong Zhang

Ph.D. Student in Statistics and Machine Learning, Georgia State University

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EDUCATION

Ph.D. in Machine Learning & Data Science **Aug. 2025 – Present**
Department of Mathematics & Statistics / Georgia State University *Atlanta, GA*

M.S. in Statistics **Aug. 2023 – May 2025**
Department of Statistics & Probability / Michigan State University *East Lansing, MI*

Bachelor of Financial Mathematics **Sep. 2019 – Jul. 2023**
School of Statistics / University of International Business and Economics *Beijing, China*

RESEARCH PROFILE

- My research interests center on statistical machine learning, reliable evaluation of generative AI, and inference for dependent and high-dimensional data.
- Methodologically, I am interested in human-aligned LLM evaluation, positive–unlabeled learning, functional data analysis and time series, and graph- and sequence-based learning, particularly for biomedical applications.

PUBLICATIONS & MANUSCRIPTS

- **Zilong Zhang**, Yi-Ting Hung, Weiyi He, Junxi Zhang, Lei Ding, Chi-Kuang Yeh. "AURA: Adaptive Uncertainty-aware Refinement for LLM-as-a-Judge Auditing" *Under review, arXiv:2606.19714*, 2026.
- **Zilong Zhang**, Yi-Ting Hung, Lei Ding, Chi-Kuang Yeh. "Quantifying and Auditing LLM Evaluation via Positive–Unlabeled Learning" *Under review, arXiv:2606.19057*, **2026**.
- Xuan Kan, Zimu Li, Hejie Cui, Yue Yu, Ran Xu, Shaojun Yu, **Zilong Zhang**, Ying Guo, Carl Yang. "R-Mixup: Riemannian Mixup for Biological Networks" *Proceedings of SIGKDD Conference on Knowledge Discovery and Data Mining 2023*, **SIGKDD 2023**.
- Xuan Kan, Wei Dai, Hejie Cui, **Zilong Zhang**, Ying Guo, Carl Yang. "Brain Network Transformer" *Proceedings of the Conference on Neural Information Processing Systems 2022*, **NeurIPS 2022**.

RESEARCH AND OTHER PROJECTS

Functional Data Analysis for Long-Run Dependence **Sept. 2025 – Present**
- *Project leaders: Dr. Chi-Kuang Yeh (Georgia State University)*
- Research on bandwidth selection for functional time series and long-run covariance estimation.

mRNA to Protein Prediction based on Transformer and GNN **Jan. 2024 – Jan. 2026**
- *Project Leaders: Dr. Yuying Xie (Michigan State University) and Dr. Lulu Shang (MD Anderson Cancer Center)*
- Developed a Transformer and graph-neural-network approach to predict protein expression from single-cell RNA measurements, combining contextual representations with graph-based modeling.
- Collaborated with researchers at Michigan State University and MD Anderson Cancer Center on a cross-institution biomedical ML project, connecting RNA-level inputs with protein-level outcomes.

RESEARCH AND OTHER PROJECTS (CONTINUED)

CRIC: Genome-Wide Association Study of Treatment-Resistant Hypertension **Sept. 2024 – Dec. 2024**

- *Project leader: Dr. Ling Wang (Michigan State University) and Dr. Yuehua Cui (Michigan State University)*
- Analyzed high-dimensional genetic data in a genome-wide association study of apparent treatment-resistant hypertension involving CKD and ARTH patient cohorts.
- Contributed statistical analyses aimed at identifying genetic variants associated with treatment-resistant hypertension; this was an exploratory research project, not a clinical finding.

Theoretical Research and Application of High Dimensional Variable Screening **Oct. 2022 – May. 2023**

- *Project leader: Dr. Wei Xiong (University of International Business and Economics)*
- Studied theoretical properties and applications of high-dimensional variable screening, where the goal is to reduce a large candidate feature set before downstream modeling.
- Compared univariate and multivariate screening approaches for dimension reduction in neural-network applications.

WORK EXPERIENCE

China Securities Co., Ltd. **Jul. 2021 – Nov. 2021**

Investment Management Intern *Beijing, China*

- Supported listing-related work for Beijing Tianzhihang Medical Technology Co., Ltd. and Golden Sun Grain and Oil Co., Ltd., contributing research and documentation across two client engagements.
- Prepared industry survey reports, risk assessments, listing application materials, market research, and internal review.
- Plan and execute fundraising activities, coordinating supporting materials and research used in the review process.

Chuzhou Longyun Financial Consulting Co. Ltd. **Mar. 2021 – Jun. 2021**

General Manager Assistant *Chuzhou, China*

- Provided recurring financial-audit and accounting support to clients, organizing financial records and documentation.
- Assisted with corporate audits and due diligence, examining financial information and preparing supporting analyses.

Bank of Communications Chuzhou Branch **Dec. 2020 – Feb. 2021**

Account Manager Intern *Chuzhou, China*

- Supported accounting-risk identification and monitoring in branch operations.
- Maintained credit databases and prepared creditworthiness assessments and analysis reports to support account reviews.

TECHNICAL SKILLS

Programming: Python, R, C; PyTorch, TensorFlow, scikit-learn, Hugging Face Transformers
Research methods: LLM evaluation, positive-unlabeled learning, partial optimal transport, uncertainty estimation, GNNs
Tools: Git, Linux, Jupyter, VS Code

HONORS & AWARDS

- **Second Prize**, Beijing Sub-Region of the 11th National University Students E-commerce "Innovation, Creativity and Entrepreneurship" Challenge Jun. 2021
- **Silver Award**, Challenge Cup Capital University Student Entrepreneurship Competition (Beijing) Sep. 2020

PROFESSIONAL MEMBERSHIPS

- Student Member, **International Chinese Statistical Association(ICSA)** 2026–Present